

APEX

Enhanced SPY Strategy

Whitepaper · Confidential Strategy Manual · Benchmark: SPY Total Return



TOTAL RETURN (20Y)	ANNUALIZED RETURN	SHARPE RATIO	MAX DRAWDOWN	BACKTEST WINDOW
2,308.7% vs SPY 662.9%	16.72% vs SPY 10.77%	0.850 vs SPY 0.624	-40.81% vs SPY -55.19%	Mar 2006–Mar 2026 20 years · 1,043 weeks

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STRATEGY THESIS

APEX is a rules-based U.S. large-cap equity program designed to improve the *ownership structure* of SPY rather than replace SPY with discretionary forecasting or concentrated stock selection. The strategy integrates four engines into one deterministic process: continuous SPY ownership, regime-sensitive leverage, a weekly **XSP** covered-call overwrite, and a midweek rebound-risk roll rule that widens the overwrite strike when early-week weakness persists. Over the 20-year March 2006–March 2026 window, a \$1M notional position compounded to **\$24.09M** under Integrated APEX versus **\$7.63M** under passive SPY buy-and-hold — delivering **+1,645.8 percentage points** of cumulative outperformance with materially reduced maximum drawdown. The edge comes from applying familiar institutional tools — leverage, overwrite, blackout discipline — in a coherent, daily-decidable process.

- **Four engines, one rulebook.** Continuous SPY ownership; regime-sensitive leverage (0.50x–1.75x); weekly XSP covered-call overwrite (7-DTE, delta-targeted); midweek roll rule. No external sleeves.
- **Regime-adaptive leverage.** Four-regime MA50/MA200 framework drives target leverage from 0.50x in bear structure (R0) to 1.75x in strong trend (R3). Rebalance only on >3% deviation.
- **Midweek roll, signal-triggered.** If intraday SPY return from 9:35 AM to 3:55 PM is $\leq -1.0\%$ and the 3:55 PM close sits in the bottom 15% of range, roll the full overwrite from 2% OTM to 3% OTM.
- **XSP as the overwrite vehicle.** Mini-SPX options are European-style, cash-settled, and taxed 60/40 under IRC §1256. Eliminates early-assignment risk and improves after-tax economics versus SPY call writing.
- **Blackout discipline.** No new calls within 3 calendar days of the second Wednesday (FOMC proxy) or second Friday (CPI proxy) of each month. Hard stop — not advisory.
- **Built for automation.** Deterministic decision engine suitable for a Finnhub-driven dashboard — phased rollout from display-only to action engine to broker-connected automation.

SECTION 1

Long-Horizon Performance

SPY buy-and-hold vs Integrated APEX · All horizons end March 10, 2026

The table below compares passive SPY total-return buy-and-hold against the final Integrated APEX strategy (with the midweek roll rule embedded inside the overwrite engine). Every horizon terminates on the same date to remove sequencing artifacts.

HORIZON	SPY BUY & HOLD	INTEGRATED APEX	OUTPERFORMANCE
1 Year	22.2%	36.7%	+14.5 pts
3 Years	82.5%	133.3%	+50.8 pts
5 Years	86.2%	141.4%	+55.2 pts
10 Years	300.4%	638.8%	+338.4 pts
15 Years	581.2%	1,300.7%	+719.5 pts
20 Years	662.9%	2,308.7%	+1,645.8 pts

SPY 20Y ENDING VALUE

\$7.63M

on \$1M notional

INTEGRATED APEX
ENDING

\$24.09M

on \$1M notional

TERMINAL WEALTH
MULTIPLE

3.16x

vs SPY buy-and-hold

ANNUALIZED SPREAD

+5.95 pts

16.72% vs 10.77%

KEY INSIGHT

Integrated APEX delivers approximately **3.16x the terminal wealth** of passive SPY on the 20-year horizon from broadly the same underlying asset exposure. The outperformance is structural, not stylistic — it comes from continuous premium harvest on the XSP overwrite sleeve, deliberate leverage calibration to market regime, and the midweek roll's protection of upside participation on rebound-prone weeks.

SECTION 2

Objective & Design Principles

Enhanced core allocation · Transparent · Repeatable · Tax-efficient

APEX seeks to deliver superior after-financing compound return relative to passive SPY buy-and-hold while preserving transparency, repeatability, tax efficiency, and disciplined risk control. It is meant to function as an **enhanced core allocation** — not a tactical trading sleeve.

APEX is designed around a simple premise: SPY is already a strong long-duration compounding asset, but passive ownership leaves structural efficiencies untapped. The strategy improves that ownership structure by allocating more capital when market structure is stronger, less capital when market structure is weaker, and harvesting XSP option premium in a controlled way from a portion of the position.

The framework avoids discretionary macro forecasting. Its signals come from observable market structure (SPY versus MA50/MA200), implied volatility (VIX), and pre-defined calendar risk controls (FOMC, CPI). That makes it suitable for eventual rules automation and daily website implementation.

Three Structural Weaknesses of Passive SPY Ownership

- **Static capital deployment** across very different market regimes — solved by regime-sensitive leverage bands.
- **No monetization of short-dated option premium** — solved by the weekly XSP overwrite.
- **Full participation in major bear-market drawdowns** — mitigated by leverage reduction in R0/R1 regimes plus overwrite income cushion.

SECTION 3

XSP as the Covered-Call Vehicle

European-style · Cash-settled · IRC §1256 tax treatment

The overwrite is implemented using **XSP (Mini-SPX Index Options)**, not SPY options. This is a deliberate structural choice for a taxable-account implementation.

ATTRIBUTE	XSP (MINI-SPX)	SPY OPTIONS
Underlying	1/10th S&P 500 Index	SPY ETF
Exercise style	European — exercise only at expiry	American — early exercise possible
Settlement	Cash settlement	Physical delivery of SPY shares
Tax treatment	IRC §1256 — 60% LT / 40% ST	Short-term ordinary income on premium
Early-assignment risk	None	Present, especially around ex-dividend
Holding-period risk on SPY	None — overlay never touches SPY lots	Assignment can force share sale

Why This Matters Operationally

APEX holds SPY shares continuously as long-duration core exposure. Writing calls directly on SPY would introduce (a) early-assignment risk around dividends, and (b) tax friction from short-term premium income offsetting long-term share lots. Writing XSP calls decouples the overlay from the underlying share lots entirely: XSP is cash-settled, cannot force an SPY share sale, and receives §1256 mark-to-market treatment with favorable 60/40 character.

Correlation, Sizing, and Implementation Notes

XSP tracks SPX directly; SPY tracks SPX at roughly 1/10th scale. Hedging a SPY share position with XSP calls is therefore structurally sound. Notional conversion is straightforward: 1 XSP contract represents approximately \$100 × SPX level of notional, which corresponds to roughly 10 SPY shares of equivalent delta exposure. The rules engine converts SPY share notional to XSP contract count as part of the Monday execution step.

- All rules previously referencing SPY weekly calls now reference **XSP weeklies**.

- 7-DTE expiry, 0.10–0.40 delta range, VIX × 1.10 IV assumption, and Monday 9:35–9:45 ET execution window carry over unchanged.
- XSP liquidity is generally adequate for institutional-scale weekly overwrite; limit orders at mid-market remain the standard.
- Assignment-prevention logic previously needed on SPY is structurally obsolete.

TAX EFFICIENCY NOTE

The §1256 treatment is not incidental to the strategy — it is a material improvement in after-tax economics. Weekly premium income that would otherwise be taxed as short-term ordinary is instead taxed 60% long-term / 40% short-term, a meaningful improvement for high-bracket taxable investors. Roll mechanics (buyback + new sale) create offsetting §1256 gain/loss pairs within the same calendar year, further reducing net taxable-income volatility.

SECTION 4

Core Decision Framework & Rule Hierarchy

A hierarchy, not a flat checklist · Earlier hard stops always prevail

APEX runs as a hierarchy, not a flat checklist. A later rule never overrides an earlier hard stop. The blackout rule, in particular, halts the new-call workflow before any other input is evaluated.

#	RULE	FUNCTION
1	Regime Classification	Establishes regime R0/R1/R2/R3 from SPY vs MA50/MA200.
2	Dynamic Leverage	Sets target leverage from regime (0.50x–1.75x).
3	Overwrite Sizing	Sets overwrite % from regime band × trend suppression.
4	Blackout Rule	Hard stop for FOMC/CPI proximity.
5	Monday Execution Protocol	8-step sequence for the weekly cycle.
6	Weekly XSP Call Construction	7-DTE, delta-targeted, VIX-adjusted.
7	Early Roll Rule	Buy back at 70% premium capture.
8	Midweek Rebound-Risk Roll	2% OTM → 3% OTM on signal.
9	Daily Monitoring	Ongoing state evaluation and action output.
10	Margin Philosophy	Financing, not recklessness; bounded and daily-costed.
11	Tax & Assignment	§1256 treatment; LT holding on SPY shares preserved.
12	Risk Controls	Market risk, model risk, execution risk disclosures.

SECTION 5

Regime Classification & Leverage

Rules 1 & 2 · The top of the decision tree · Every downstream rule depends on it

The four-regime framework sits at the top of the decision engine. Every downstream rule inherits from it — leverage, overwrite %, delta targeting, and rebalance logic all chain off the regime classification.

REGIME	CONDITION	TARGET LEV.	BASE Δ	VIX ADJ. (LOW / HIGH)	OW% BAND
R0 — Bear	Below MA50 and MA200	0.50x	0.15	0.20 / 0.10	100%
R1 — Weak	Below MA50, above MA200	1.25x	0.20	0.25 / 0.15	25–100%
R2 — Mixed	Above MA50, below MA200	1.25x	0.25	0.30 / 0.20	25–100%
R3 — Strong	Above MA50 and MA200	1.75x	0.30	0.35 / 0.25	25–100%

VIX Adjustment Ladder

When **VIX** < 16, use the higher delta in each row (reflecting a low-premium environment that justifies tighter strikes). When **VIX** > 25, use the lower delta (reflecting elevated premium that compensates for wider strikes). Between 16 and 25, use the base delta.

Leverage Rebalancing

Compute target leverage from regime. Compare to actual leverage after current SPY price and account equity update. If absolute deviation exceeds **3%**, emit a buy/sell SPY instruction at Monday open. Display current financed notional and estimated financing drag on the dashboard daily.

SECTION 6

Overwrite, Blackout & Execution

Rules 3–6 · Daily operational mechanics of the overwrite sleeve

Rule 3 — Overwrite Sizing

- Regime sets the **permitted** overwrite band.
- Trend-suppression score then **collapses** the band to 25%, 50%, or 100%: trend suppression > 0.70 → 25% · 0.50–0.70 → 50% · < 0.50 → 100%.
- Blackout suppresses any new overwrite regardless of percentage.
- Midweek roll widens strike but does **not** change overwrite percentage.
- Exact trend-suppression formula is an admin-set parameter until locked; the rules engine exposes it as a calculated field.

Rule 4 — Blackout Rule (Hard Stop)

No new calls are opened within **3 calendar days** of:

- The **second Wednesday** of each month (FOMC proxy), or
- The **second Friday** of each month (CPI proxy).

When blackout = TRUE, all new call-sale instructions are suppressed for that cycle. The rule applies *before* delta selection and strike placement — no later rule can override it.

Rule 5 — Monday Morning Execution Protocol

- 1 Data pull.** Fetch SPY, VIX, MA50, MA200 from the market-data feed (Finnhub).
- 2 Regime check.** Classify as R0/R1/R2/R3; display in dashboard header.
- 3 Blackout check.** If TRUE, stop — no new calls this cycle.
- 4 Rebalance check.** If $|\text{target} - \text{actual leverage}| > 3\%$, emit buy/sell SPY at market open.
- 5 Early-roll check.** If existing call has captured $\geq 70\%$ of premium, buy it back.
- 6 Sell new XSP call.** Black-Scholes delta targeting, VIX adjustment, trend suppression; limit order near mid, 9:35–9:45 ET.
- 7 Log trade.** Strike, expiry, premium, contracts, OW%, regime, leverage, portfolio value.
- 8 Monitor.** Live roll-progress tracking surfaced on the dashboard each day.

Rule 6 — Weekly XSP Covered-Call Construction

PARAMETER	RULE	NOTE
Underlying	XSP (Mini-SPX Index Options)	European, cash-settled, \$1256
Expiry	Weekly 7-DTE	Sold in Monday window
Execution window	Monday 9:35–9:45 AM ET	Limit near mid-market
Delta	0.10–0.40, VIX-adjusted	Derived from regime + VIX ladder
IV assumption	VIX $\times$ 1.10	Used in B-S strike targeting
Order type	Limit at mid	Avoid market orders
Base strike	$\approx$ 2% OTM	Subject to midweek roll to 3% OTM
Assignment policy	N/A — XSP is European, cash-settled	No underlying shares at risk

Indicative Annualized Premium Yield Bands

These are environment expectation bands, not promises. Realized yield varies with regime and VIX regime transitions.

- **Low VIX (< 16):** approximately **5–8%** annualized.
- **Medium VIX (16–25):** approximately **8–14%** annualized.
- **High VIX (> 25):** approximately **14–22%** annualized.

SECTION 7

The Midweek Rebound-Risk Roll Rule

Rules 7–8 · The integrated enhancement · 2% OTM → 3% OTM on signal

Rule 7 — Early Roll Rule (Base)

Before the midweek rule is evaluated, the base early-roll rule already monitors the open XSP call daily. When captured premium reaches **70% or greater** — computed as $(\text{premium_received} - \text{current_call_mark}) / \text{premium_received}$ — the system emits a buyback instruction and proceeds to the new-sale logic (subject to blackout and other hard stops).

Rule 8 — Midweek Rebound-Risk Roll Rule

The new enhancement activates only *after* a base overwrite is already in place. Research first showed that a separate short-term overlay trade was not capital-efficient enough to justify a standalone sleeve. The signal then proved more useful as an *overwrite-management* rule. Multiple variants were tested — direct overlays, overwrite reduction, strike widening, midweek roll simulation against the weekly overwrite structure. The winning implementation was a **full roll from 2% OTM to 3% OTM**.

STEP	OPERATING DEFINITION
Base overwrite	Opens each Monday at ≈2% OTM with a 5-trading-day horizon.
Signal window	Monday, Tuesday, Wednesday — through 3:55 PM ET.
Condition 1	SPY intraday return from 9:35 AM to 3:55 PM is $\leq -1.0\%$.
Condition 2	The 3:55 PM close sits in the bottom 15% of that day's intraday range.
Action	If both conditions trigger, roll the full overwrite from 2% OTM to 3% OTM, same remaining weekly expiry.
Do NOT	Change leverage. Change overwrite percentage. Add positions.

Testing & Research Path

The rule was tested across **939 synthetic weekly cycles** and triggered **157 times** (~16.7% of cycles). Total synthetic benefit of **+\$50.93 per share** before mapping through the workbook's actual overwrite return path. Full-roll implementation dominated partial-roll. 3% OTM target dominated 4% OTM target. The rule's effect was strongest when weakness persisted into Tuesday and Wednesday — consistent with the rebound logic. Final integrated horizon figures map the rule through the workbook's actual XSP overwrite return path, not an arbitrary sleeve assumption.

RULE RATIONALE

The Signal Identifies Rebound-Prone Weeks

Research Note — APEX Strategy Development

"The signal does not say to stop overwriting. It says that tight calls become less attractive during a specific setup: early-week persistent weakness with the close near the intraday lows. In those weeks, widening the strike reduces the probability that the overwrite will unnecessarily cap a rebound."

Trigger Profile: 157 triggered rolls across 939 synthetic weekly cycles · Total synthetic benefit +\$50.93 per share before being mapped through the workbook's actual XSP overwrite return path.

SECTION 8

Daily Monitoring & Action States

Rule 9 · Minimum daily checks · Deterministic action output

Outside the Monday opening workflow, the website evaluates open-overwrite state each trading day. The daily monitor outputs **exactly one action state** after running the minimum daily checks.

Minimum Daily Checks

- Current regime (may have changed intraweek from the Monday classification).
- Blackout status for any upcoming new-call cycle.
- Current short-call mark-to-market and premium-capture percentage.
- Midweek roll signal status (Monday, Tuesday, Wednesday only).

Daily Monitor Action States

ACTION STATE	TRIGGER
HOLD	Default — all systems nominal, no rule triggered.
REBALANCE SHARES	Monday check: $ \text{target} - \text{actual leverage} > 3\%$.
BUY BACK CALL	Early-roll trigger: captured premium $\geq 70\%$.
SELL NEW CALL	After buyback, or fresh Monday cycle with no blackout.
ROLL CALL TO 3% OTM	Midweek signal triggered (Mon/Tue/Wed).
NO NEW CALLS	Blackout window active (FOMC or CPI proximity).

SECTION 9

Margin, Tax & Risk

Rules 10–12 · Capital-efficiency philosophy, tax posture, and honest risk disclosure

Rule 10 — Margin & Capital-Efficiency Philosophy

The strategy treats margin as **financing, not recklessness**. Risk depends on the asset financed, the amount of leverage used, and the rules governing its use. Prudent index-level financing on a diversified, liquid benchmark at bounded leverage (max 1.75x) is analytically reasonable — but financing cost must be recognized daily, and aggregate balance-sheet exposure matters. The dashboard displays current leverage, target leverage, financed notional, and estimated financing line at all times.

Rule 11 — Tax & Assignment Considerations

- Underlying SPY shares are intended to remain unsold, accruing **unrealized long-term capital gain**.
- XSP premiums receive IRC §1256 treatment: 60% long-term / 40% short-term capital gain**, marked-to-market at year-end. This is a material improvement versus SPY call writing, where premium would be ordinary short-term income.
- Roll mechanics (buyback + new sale) create offsetting §1256 gain/loss pairs, reducing net taxable-income volatility.
- Assignment risk on the underlying SPY shares is structurally **eliminated** by using XSP (European, cash-settled) rather than SPY options.
- The website logs each overlay action separately from the underlying share-lot register.

Consult a qualified tax advisor on §1256 treatment for specific account structures and state-level considerations.

Rule 12 — Risk Controls & Limitations

- APEX remains fully exposed to broad equity market risk; continuous SPY ownership is preserved by design.
- Leverage amplifies losses. Bounded leverage (max 1.75x) and regime-based reduction to 0.50x in R0 are the primary mitigants.
- Fast drops before moving-average confirmation (gap-down regime transitions) can still hurt.
- Model risk: Black-Scholes with $IV = VIX \times 1.10$ is an approximation, not a true implied surface.
- Bid/ask frictions are not fully modeled; live execution slippage must be monitored against backtest assumptions.
- Weekly overwrite reduces upside in strong bull phases; the 25/50/100% overwrite banding is the primary mitigant.
- The midweek roll rule is a **strike-management improvement**, not a guarantee against missed upside.

SECTION 10

Website Rules Engine Specification

Deterministic decision engine · Phased rollout · Finnhub-driven

The strategy is represented as a deterministic decision engine. The engine ingests a daily market snapshot and returns exact instructions. The engine maintains a **daily decision log**: timestamp, input snapshot, rule state, recommended action, and execution confirmation — material for governance, debugging, and future walk-forward validation.

CATEGORY	DESCRIPTION	EXAMPLE FIELDS
Market inputs	Live prices, trend state, volatility, macro calendar	spy_last, spy_0935, spy_1555, vix, ma50, ma200, cpi_blackout, fomc_blackout
Portfolio inputs	Current strategy state	shares, leverage_actual, leverage_target, xsp_call_strike, xsp_call_expiry, premium_received, call_mark, overwrite_pct
Derived fields	Rule-engine computations	regime, trend_suppression_score, target_delta, premium_capture_pct, signal_flag
Action outputs	Exact instructions for operator or API	rebalance_shares, buyback_call, sell_new_call, roll_call, hold

Phased Rollout

- Phase 1 — Display-only dashboard.** Regime, target leverage, overwrite %, target delta, blackout status, midweek roll signal state. No instructions emitted.
- Phase 2 — Action engine.** Emit exact instructions each Monday and at each midweek check.
- Phase 3 — Trade logging and historical rule-state archive.** Full audit trail of decisions and executions.
- Phase 4 — Broker-connected automation.** Only after Phases 1–3 prove stable across multiple regime transitions.

SECTION 11

Conclusion

One integrated enhanced-SPY strategy · One rulebook · One daily decision

APEX is now a single integrated enhanced-SPY strategy with a clear rule hierarchy: determine regime, set target leverage, determine overwrite size, enforce blackout and early-roll rules, sell the weekly XSP call, and widen the strike to 3% OTM if the midweek rebound-risk signal appears. The value of the midweek roll rule is not that it creates a second strategy. Its value is that it improves the existing overwrite engine at the precise point where tight calls are most exposed to rebound risk.

This document is intended to be sufficiently explicit that the strategy can be implemented manually, reviewed by an investment committee, or translated into a Finnhub-driven website without ambiguity in the daily decision process.

THE CORE CONVICTION

APEX is a **more efficient ownership structure for SPY** — not a substitute for it. Four engines in sequence, one rulebook, one daily decision. XSP as the overwrite vehicle eliminates early-assignment risk and improves after-tax economics through §1256 treatment. The 20-year record — **\$24.09M versus \$7.63M** for passive SPY on \$1M notional, with lower maximum drawdown and a higher Sharpe ratio — argues the discipline is worth the operational cost.

IMPORTANT DISCLOSURES

Disclosures & Legal Notice

Backtest caveat · Risk disclosure · Tax notice · Distribution restriction

STRATEGY DESCRIPTION

APEX is a rules-based investment strategy. This manual describes its operating rules, backtest results, and implementation framework as of the publication date. It reflects the final integrated strategy with the midweek roll rule embedded in the weekly XSP overwrite engine.

BACKTEST CAVEAT

Backtested performance is **hypothetical, does not represent actual trading, and is prepared with the benefit of hindsight**. Backtested results do not reflect the impact of material economic and market factors on decision-making if actual capital had been managed, and future results may differ materially from the figures presented. All performance figures reflect gross-of-fees modeled returns unless otherwise stated.

RULE-TESTING METHODOLOGY

The midweek roll rule was tested across 939 synthetic weekly cycles using the workbook's actual overwrite return path combined with synthetic option logic for the roll decision. Live execution may diverge from synthetic results due to bid/ask frictions, fill quality, and timing latency.

PERFORMANCE COMPARISON METHODOLOGY

Sharpe ratio and maximum drawdown figures on the cover are drawn from the Mar 2006–Mar 2026 institutional backtest window and are directionally representative of the Integrated APEX profile. SPY comparison figures use SPY total-return (dividend-reinvested) data over the same window.

RISKS

Investing in equity and option strategies involves significant risk of loss, including **loss of principal**. Leverage magnifies both gains and losses. Option writing caps upside and does not eliminate downside. Rules-based strategies can underperform in regime transitions, choppy markets, and gap events. Past performance is not indicative of future results.

TAX DISCLOSURE

References to IRC §1256 treatment of XSP options reflect general federal tax characterization as of the publication date and are not tax advice. Individual tax outcomes depend on account type, state of residence, entity structure, and other facts. **Consult a qualified tax advisor before implementation.**

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